TOPIC 3

Why Stochastic Gradient Descent: Explained

The original class slides, with a plain explanation under each one

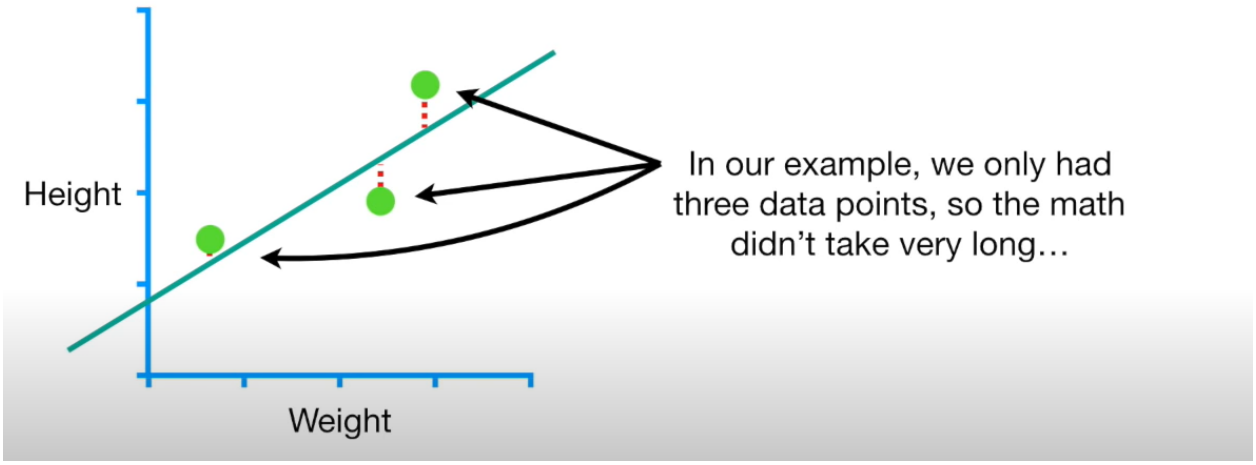
> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

Gradient descent as shown so far looks at **every data point** at every step. These three slides explain why that becomes a problem for big data sets and what **stochastic gradient descent** does about it. (The next topic goes deeper.)

Words you will meet

Word	Plain meaning
Data point	One row of the data, here one person's weight and height
Stochastic	Involving randomness
Stochastic gradient descent (SGD)	Gradient descent that uses a randomly chosen small part of the data at each step

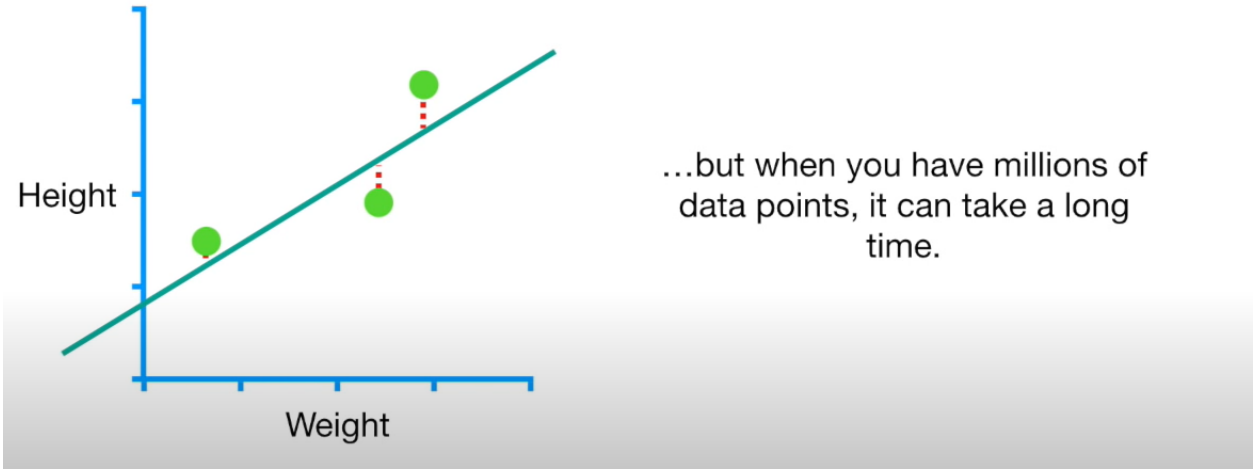
Slide 1: Small data: the maths is quick



Slide 1

In our example there were only **three data points** (the green dots; the dotted red lines are their residuals). Every step of gradient descent had to add up three terms, so the calculation was quick.

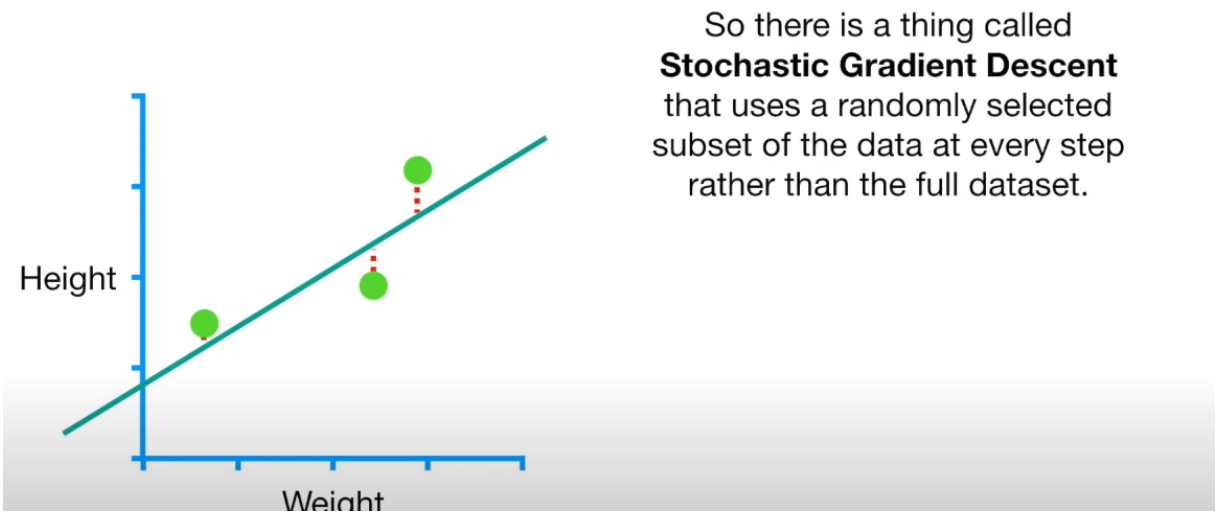
Slide 2: Big data: the maths takes a long time



Slide 2

The same picture, but the message is: when you have **millions of data points**, adding up a term for every one of them at every step can take a very long time. Gradient descent as we have learned it does that.

Slide 3: The fix: Stochastic Gradient Descent



Slide 3

Stochastic Gradient Descent (SGD) uses a **randomly selected subset of the data at every step** instead of the full data set. Each step is therefore much cheaper to compute, and with many cheap steps you still head towards the minimum.

- Full data at every step: accurate but slow on big data
- Random subset at every step: fast, slightly noisy, usually good enough